CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of 5 numbers is 30. When one number is removed the mean of the remaining 4 becomes 27. The removed number is: 1

(a) 42 **(b)** 30 **(c)** 3 **(d)** 15

2. The mean of 10 values was found to be 36. Later it was found that 80 was misread as 30. The correct mean is: 1

(a) 86 **(b)** 36 **(c)** 31 **(d)** 41

3. A class has 29 students with mean age 13 years. When the teacher's age is included, the mean rises to 14. The teacher's age (years) is: 1

(a) 42 **(b)** 38 **(c)** 43 **(d)** 14

4. The mean of the first 12 odd natural numbers is: 1

(a) 12 **(b)** 13 **(c)** 144 **(d)** 6.5

5. A data set 54, 34, 23, 16, 31, 46, 57 gets an extra (outlier) value 540. The median of the new data of 8 values is: 1

(a) 34 **(b)** 540 **(c)** 40 **(d)** 100.12

6. A car travels 120 km at 40 km/h and then 120 km at 60 km/h. Its average speed (km/h) for the whole journey is: 1

(a) 50 **(b)** 48 **(c)** 90 **(d)** 24

7. The mean of 8 numbers is 15. If every number is multiplied by 3 and then 4 is added, the new mean is: 1

(a) 45 **(b)** 19 **(c)** 57 **(d)** 49

8. The mean of 5 numbers is 24. When one number is removed the mean of the remaining 4 becomes 21. The removed number is: 1

(a) 36 **(b)** 3 **(c)** 24 **(d)** 9

9. **Assertion (A):** The mean of 4, 6, 8, 10 and 12 is 8. 1
Reason (R): The mean of a data set is the sum of the observations divided by the number of observations.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The median of 3, 7, 9, 12 and 15 is 9. 1
Reason (R): The mode of the data 2, 3, 3, 5 is 3.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean of 6 numbers is 20. When one number is removed, the mean of the remaining 5 numbers is 18. Find the removed number. 2

12. The mean of the first 8 even natural numbers is 9. Verify this for n = 8. 2

13. A data set was recorded as 45, 27, 21, 42, 20, 43, but the first value should have been 39. Find the correct mean. 2

14. The mean of 6 numbers is 35. When one number is removed, the mean of the remaining 5 numbers is 32. Find the removed number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A class of 40 students has mean mark 62. After rechecking, a score of 38 was corrected to 80. Find the new mean. 3

16. The mean age of 11 players is 23 years. When the coach is included the mean becomes 25. Find the age of the coach. 3

17. A student's marks in 5 tests are 49, 75, 64, 74, 48. What mark must she score in the 6th test so that her mean becomes 68? 3

OR

A class of 25 students has mean mark 48. After rechecking, a score of 20 was corrected to 70. Find the new mean. 3

18. The mean age of 15 players is 20 years. When the coach is included the mean becomes 22. Find the age of the coach. 3

OR

A student's marks in 5 tests are 51, 66, 63, 62, 68. What mark must she score in the 6th test so that her mean becomes 68? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher records the marks (out of 20) of seven students in a quiz: 9, 12, 15, 15, 10, 18, 19. 5
(i) Find the mean.
(ii) Find the median.
(iii) Find the mode.
(iv) Find the range, and the new mean if an eighth student scores 20.

20. The mean of 10 numbers is 40, the mean of 20 numbers is 50 and the mean of 20 numbers is 30. Find the mean of all 50 numbers. 5

OR

The mean of 15 observations is 20. If each observation is multiplied by 3 and then decreased by 8, find the new mean and the new sum. 5

END OF QUESTION PAPER — Total: 40 Marks